# **The Anatomy of a Successful Restaurant:** How Business Features and Engagement on Yelp Drive High Ratings

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CS 784: Data Management

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# Yelp Open Dataset

An all-purpose dataset for learning

## The Dataset

**6,990,280 reviews**

**150,346 businesses**

**200,100 pictures**

**11 metropolitan areas**

908,915 tips by 1,987,897 users
Over 1.2 million business attributes like hours, parking, availability, and ambience
Aggregated check-ins over time for each of the 131,930 businesses

# About Yelp

- 5.8 million claimed businesses
- 524,000 paying advertising locations
- 287 million reviews
- 74 million monthly visits

Yelp Blog for Businesses

BUSINESSES

## Yelp 101: a business guide to thriving on Yelp

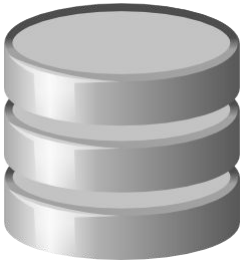
Amanda Retzer
July 27, 2022

# Problem

- Yelp provides some insights
- features - business attributes
  behaviors - engagement metrics
- higher rating = successful/popular/good
- ultimate goal: high Yelp rating

# Objective

- identify recommendations for businesses
- focus on restaurants
- for businesses to increase in popularity (higher ratings)
- and to increase customer engagement (more reviews)



# Related Work

**Hegde et al. “Restaurant Setup Business Analysis Using Yelp Dataset”**

- focuses on new businesses
- analyzes high-frequency business attributes
- uses a k-d tree algorithm to find close-by restaurants

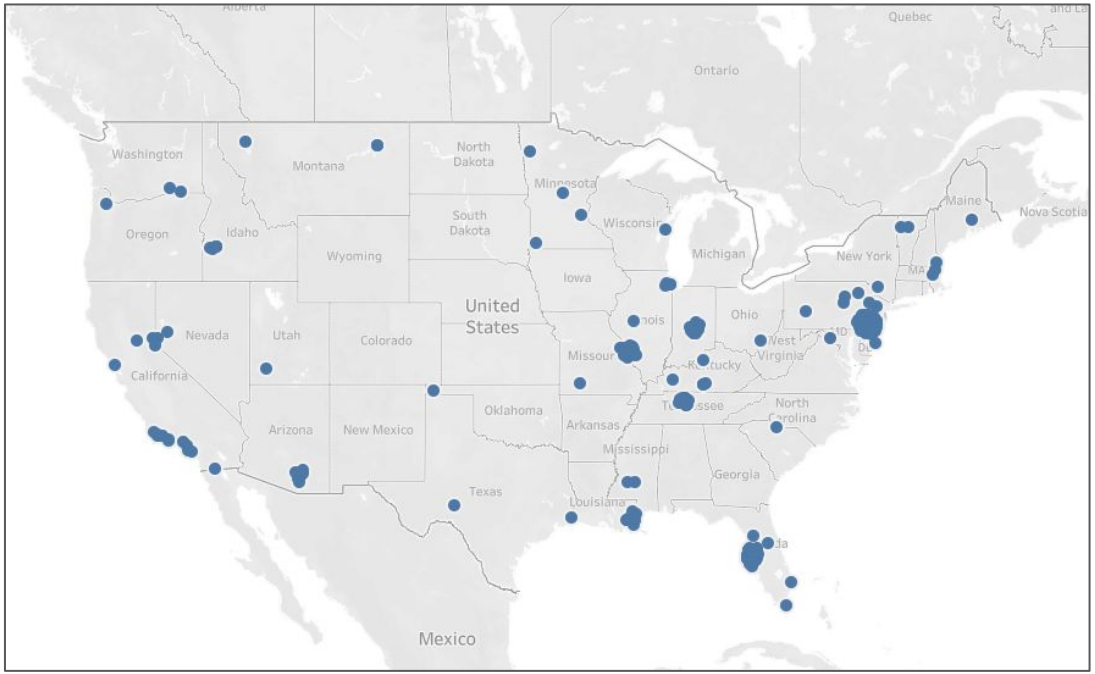
**Cervellini et al. “Finding Trendsetters on Yelp Dataset”**

- trendsetters: users who identify emerging businesses early and influence others
- focuses on tips
- user-based > business-based

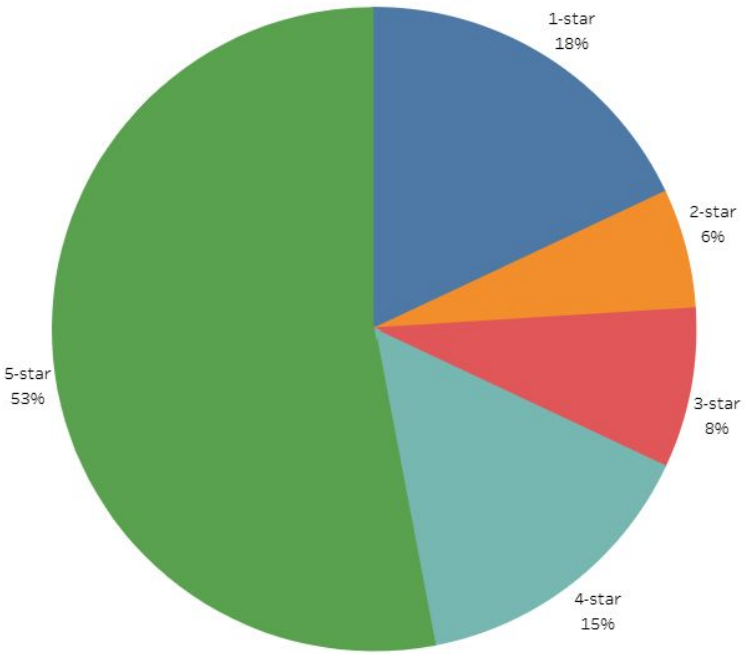
# About the Dataset

1. businesses
    - attributes
    - categories
1. checkins
2. reviews
3. tips
4. users

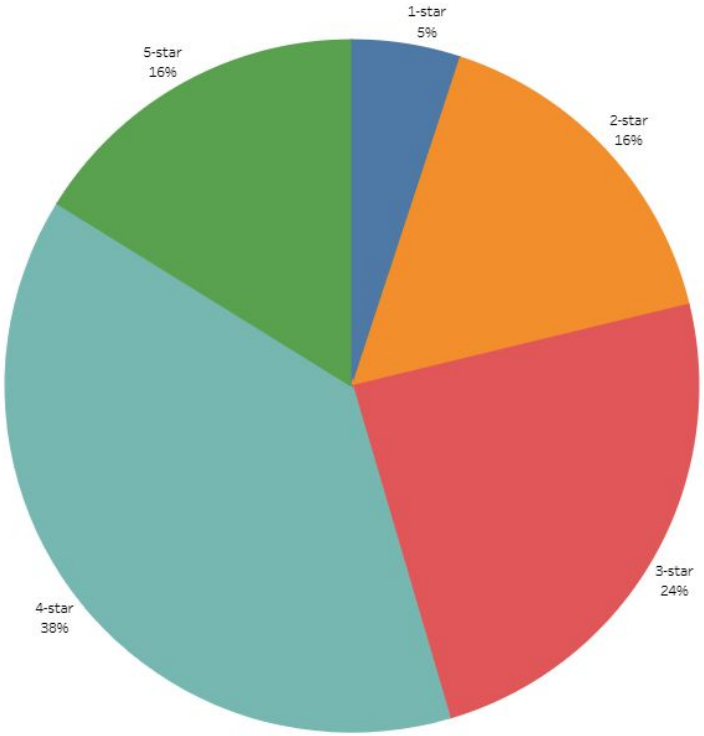
extract - not all cities are represented



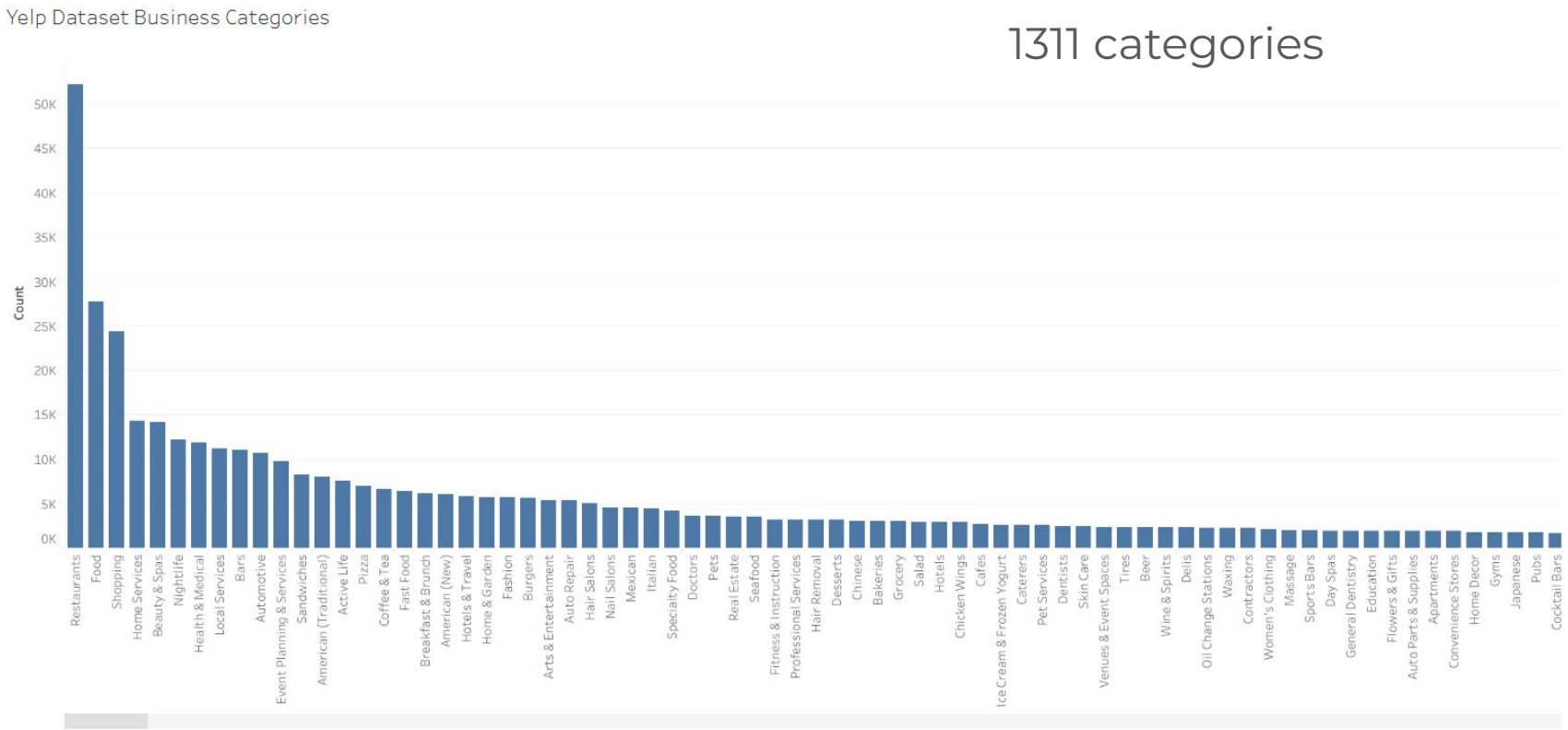
## Distribution of Review Star Ratings




## Distribution of Average Business Ratings

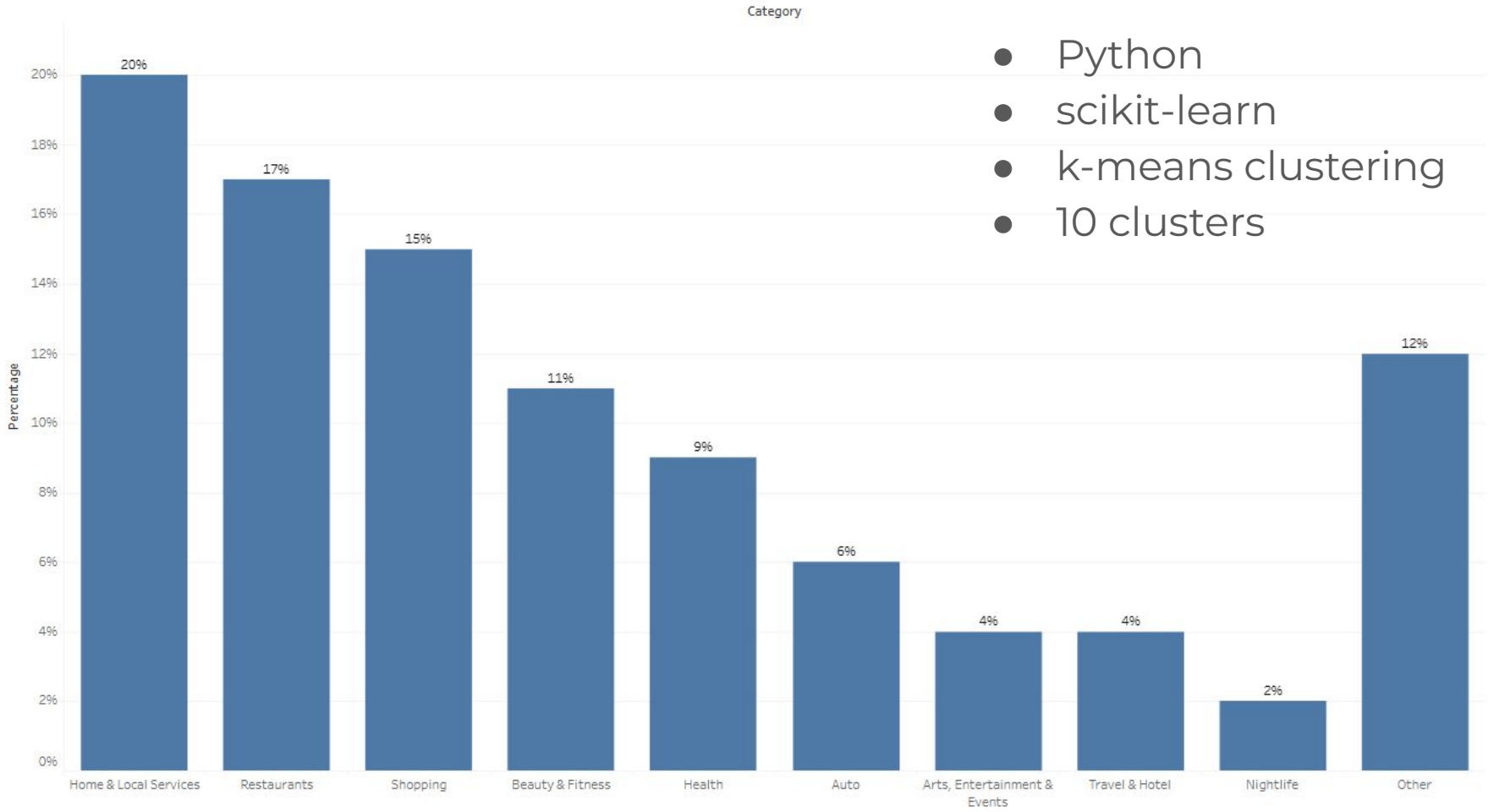


# Categories Analysis




1311 categories

Reviewed Businesses By Category




- Python
- scikit-learn
- k-means clustering
- 10 clusters

# Business Attributes

- **boolean**
  - OutdoorSeating, WheelchairAccessible, BusinessAcceptsCreditCards
- **categorical**
  - Wifi: free, no, etc.
  - NoiseLevel: average, quiet, etc.
  - RestaurantsAttire: casual, formal, etc.
- **nested**
  - BusinessParking: garage, street, validated, lot, etc.
  - Ambience: romantic, classy, trendy, etc.
- **numerical**
  - RestaurantsPriceRange: 1, 2, 3 ($, $$, $$$)

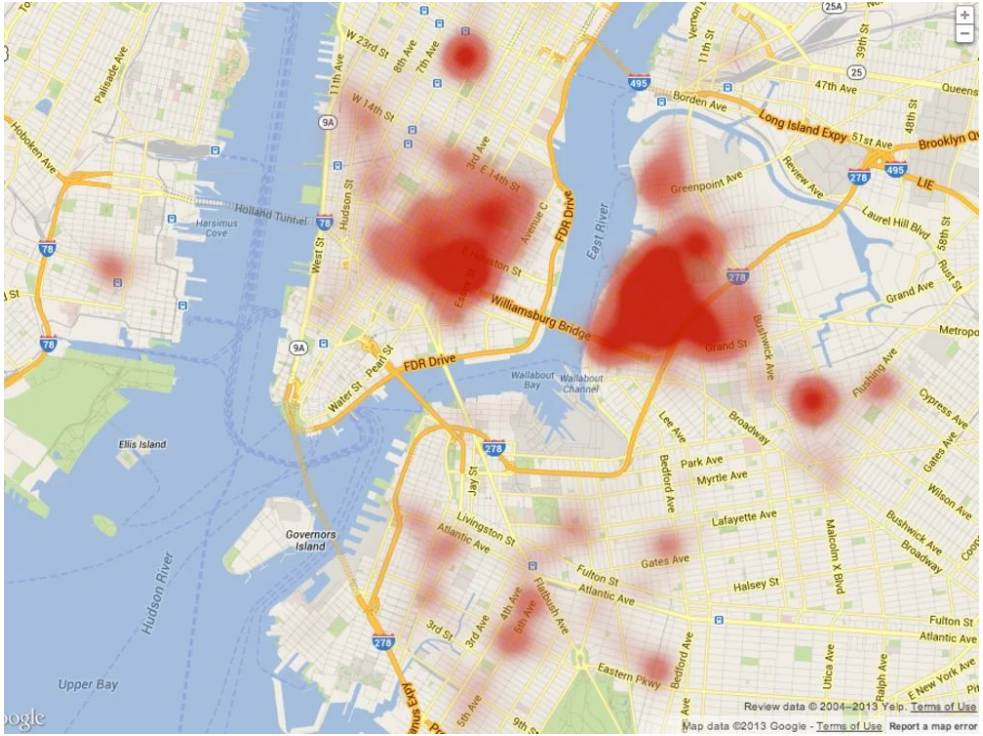
# Business Attributes Correlated to Number of Reviews and Rating

# Takeaways

- majority of reviews are 5 stars
- convenience: delivery, takeout, outdoor seating
- preferences: ambience, good for groups
- high-impact features

# Further Analysis

- temporal
- spatial
- textual




Yelp Word Map

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